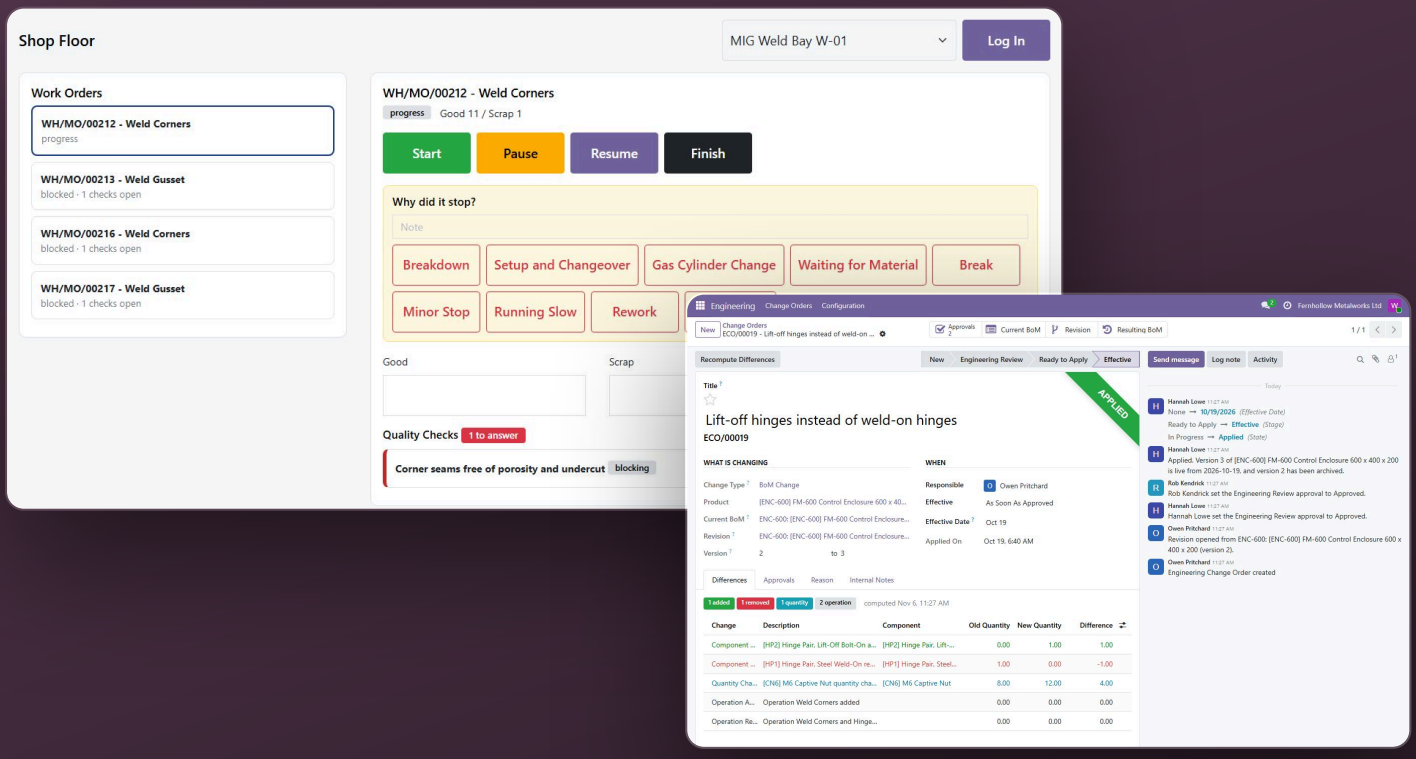
WAY4TECH SOLUTIONS

Live demo
manufacturing-demo.way4tech.com
w4demo / Demo2026 · Odoo 19, Community

COMPLETE GUIDE · SETUP, THE TERMINAL, OEE AND ENGINEERING CHANGES

Shop Floor Control and PLM for Odoo 19

An operator terminal with stop reasons, production that cannot be counted twice, quality checks that hold the line and OEE worked out from real time logs. Engineering change orders over bills of materials with an archived revision, the difference worked out, approvals per stage and versioned history. Every screen, shown on a live demo.



START About this guide

Two problems sit next to each other on every factory floor. Nobody can say where the shift went, because the machine time is guessed at the end of the day. And nobody can say which version of a product was built last Tuesday, because the bill of materials was edited in place. This guide shows how two Way4Tech modules for Odoo 19 Community answer both, one screen at a time. You do not need to have used Odoo before.

Who it is for

The production manager, the shift supervisor and the operators who use the terminal, the design and quality engineers who raise and approve changes, and anyone being shown the system for the first time.

How to read it

Front to back the first time. Each part opens with what you will learn and a short thing to try on the live demo. Every heading carries a badge saying which module it belongs to.

The live demo

Fernhollow Metalworks Ltd, an invented sheet metal shop in Rotherham with a fibre laser, a press brake, a weld bay, a powder coat line and an assembly bench. Nine weeks of production and twelve change orders. Open it and follow along.

HOW IT WORKS

The demo factory

Address	**manufacturing-demo.way4tech.com**
Sign in	**w4demo / Demo2026**
Company	Fernhollow Metalworks Ltd · five work centres · four products
Runs on	Odoo 19, Community edition, with Manufacturing

The two modules

BADGE	MODULE	TECHNICAL NAME	PRICE	NEEDS
SHOP FLOOR	Shop Floor Control	`wt_shop_floor`	USD 299	Manufacturing
PLM	PLM, Engineering Change Orders	`wt_plm_eco`	USD 149	Manufacturing

Each module works on its own. Installed together, a change order that alters an operation carries the quality checks of that operation into the new version, so the terminal asks the right questions the day the change goes live.

> Every company, person and number on the demo is invented. The demo is shared and reset from time to time, so what you see may have moved on from the screenshots.

What is in the guide

PART 1 What the shop floor and the drawing office need

PART 2 Installing, roles and the one setting

PART 3 Setting up the floor SHOP FLOOR

PART 4 The terminal, step by step SHOP FLOOR

PART 5 Work orders, sessions and the event trail SHOP FLOOR

PART 6 Quality checks that hold the line SHOP FLOOR

PART 7 Measuring OEE SHOP FLOOR

PART 8 Change types, stages and the board PLM

PART 9 Raising a change and editing the revision PLM

PART 10 Approvals PLM

PART 11 Applying, versions and traceability PLM

PART 12 Printed documents

PART 13 A fifteen minute walk-through

PART 14 Questions people ask

PART 15 The menu map

PART 1 What the shop floor and the drawing office need

Standard Odoo Manufacturing already has work centres, bills of materials, operations and work orders. What it does not have is a screen an operator can use with a glove on, a reason behind every stop, quality checks that hold the line, and a controlled way to change a bill of materials.

IN THIS PART YOU WILL LEARN

- Sessions, stops, production and checks, and why each one is recorded
- What OEE is made of
- What an engineering change order is
- Where each module starts and stops

The shift, as the terminal sees it SHOP FLOOR

One operator, one machine, one shift

1. **Log in** — A session on the work centre
2. **Start** — Productive time begins
3. **Pause** — A reason, so the minutes land in the right loss
4. **Record** — Good and scrap off the machine
5. **Checks** — Measures, pass/fail, batch numbers
6. **Finish** — Refused while a blocking check is open

Every one of those presses writes a **terminal event** and, for start, pause and resume, a real Odoo machine time log (`mrp.workcenter.productivity`). That is what makes the figures add up later: the minutes Odoo already keeps for a work centre are the same minutes the terminal recorded.

OEE in one paragraph SHOP FLOOR

RATIO	WORKED OUT AS	WHAT LOWERS IT
Availability	Run time / planned time	Breakdowns, setup, waiting for material, breaks
Performance	Ideal cycle time x units made / run time	Minor stops, running slow
Quality	Good units / units made	Scrap and rework
OEE	Availability x performance x quality	All of the above

The change order PLM

An engineering change order

1. **Raise** — What and why, against the live BoM
2. **Revision** — An archived copy to edit
3. **Differences** — Worked out line by line
4. **Approvals** — Set on the stage
5. **Apply** — Version + 1, old one archived

The live bill of materials is never edited while a change is being discussed. Manufacturing orders keep using it until the change is applied, and the archived revision can never be picked up by mistake.

What is not in either module

NOT INCLUDED	WHERE IT LIVES INSTEAD
Machine connections (PLC, counters, OPC)	Not built. Quantities are typed on the terminal.

Planning and scheduling	Standard Odoo Manufacturing
Maintenance requests from a breakdown	Not built. The breakdown note is on the event.
Drawings and document control	Not built. Attach files to the change order chatter.
Statistical process control charts	Not built. Check results can be listed and grouped.

PART 2 Installing, roles and the one setting

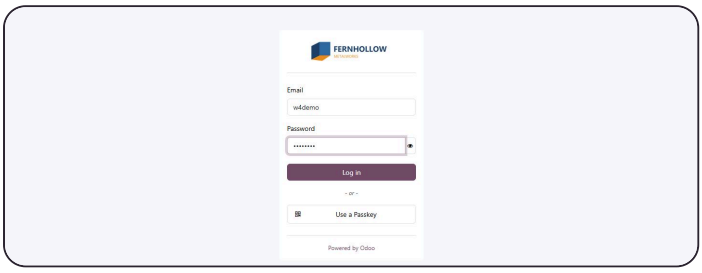
Both modules install on Odoo 19 Community with Manufacturing and nothing else. Neither adds a settings page of its own: everything is configured on records.

IN THIS PART YOU WILL LEARN

- The one Manufacturing setting that has to be on
- The four roles across the two modules
- Signing in to the demo

TRY IT YOURSELF ON THE DEMO

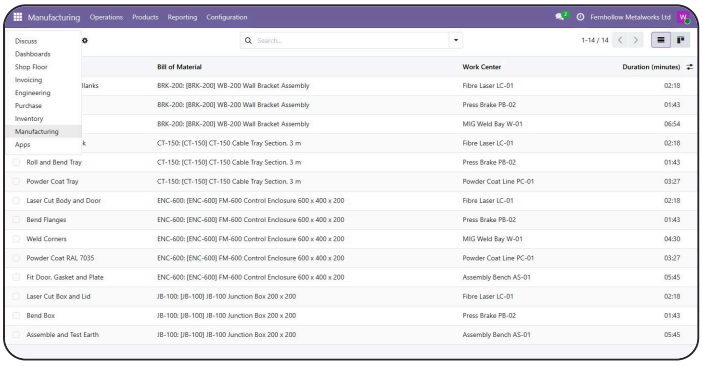
1. Sign in as w4demo and open the app menu
2. Find Shop Floor and Engineering next to Manufacturing



1

Sign in

Open the demo address and sign in with the shared login. The demo user can use every screen in this guide but cannot change system settings.



2

The apps

Shop Floor and Engineering sit beside Manufacturing and Inventory. Shop Floor appears for anyone who may operate a terminal, Engineering for anyone who may raise a change.

Setup from zero

WHERE	SETTING	WHY IT MATTERS
Manufacturing > Configuration > Settings	**Work Orders** on	Operations and work orders only exist with it. The terminal works on work orders. SHOP FLOOR
Users	Shop Floor: **Operate the Terminal**	Operators. Includes Manufacturing user rights. SHOP FLOOR
Users	Shop Floor: **Configure the Shop Floor**	Supervisors: stop reasons, checks, Release Checks, Measure OEE, reset a failed check. SHOP FLOOR
Users	Engineering Changes: **Raise Engineering Changes**	Engineers who raise and edit change orders and answer their own approvals. PLM
Users	Engineering Changes: **Configure Engineering Changes**	Change types, stages, approvers, and answering any approval. PLM
Work centre	Ideal cycle time, stop reasons offered	Performance and the pause buttons. SHOP FLOOR
Operation	Quality checks	What the terminal asks on every work order. SHOP FLOOR

Engineering > Configuration	Stages and approvers	Who has to sign before a change applies. PLM

> **Both modules on one database:** starting a revision copies the bill of materials with its operations, and the operations carry their shop floor checks. An engineer therefore also needs **Configure the Shop Floor** to start or apply a change order on a bill of materials that has checks. On the demo the engineers have it.

Every model in both modules has a multi-company rule, so a second company in the same database sees only its own sessions, checks, change orders and stages (stages and change types left without a company are shared).

PART 3 Setting up the floor SHOP FLOOR

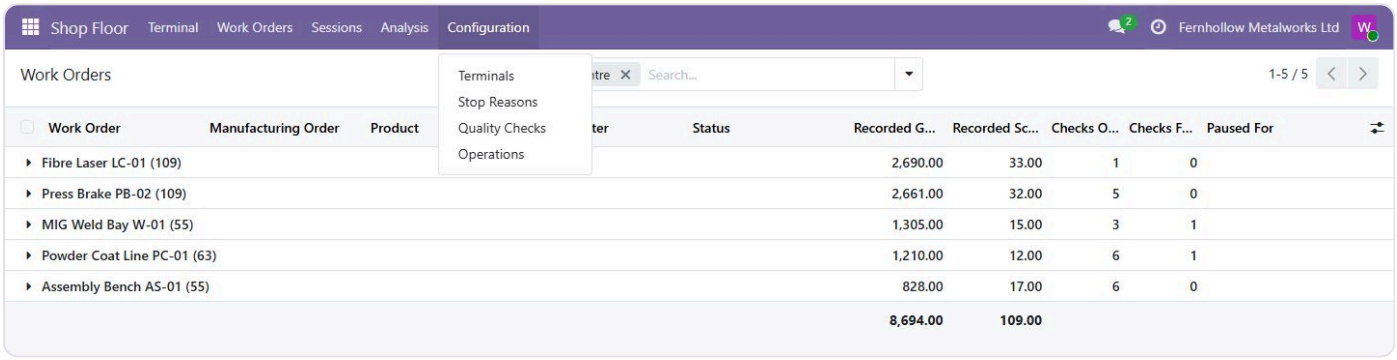
Four configuration screens decide what the terminal shows: the work centres it runs on, the reasons an operator can give for a stop, the quality checks it asks, and the operations they hang on.

IN THIS PART YOU WILL LEARN

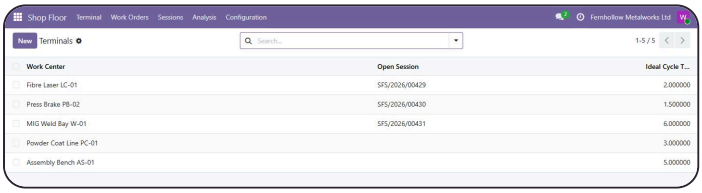
- Ideal cycle time and the stop reasons offered per machine
- Loss category, blocking, note required and expected duration on a stop reason
- Measure, pass/fail and text checks, blocking or not
- Checks that apply only to one product

TRY IT YOURSELF ON THE DEMO

1. Open Configuration > Terminals > Fibre Laser LC-01
2. Open Configuration > Stop Reasons > Breakdown
3. Open Configuration > Quality Checks > Body blank width

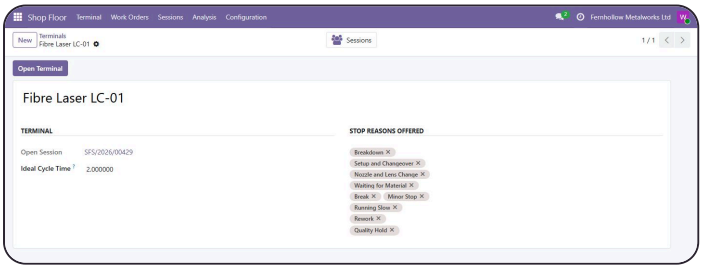


Shop Floor > Configuration: Terminals, Stop Reasons, Quality Checks, Operations.



1 Terminals

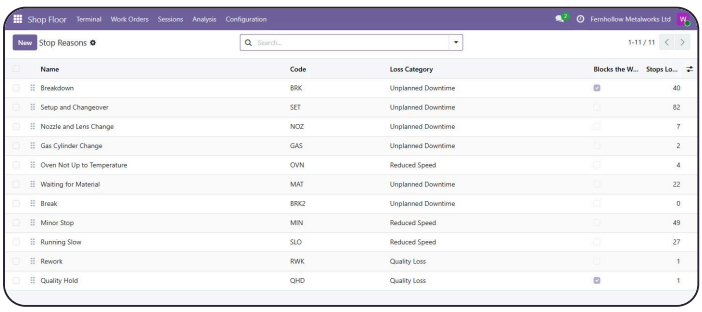
Every work centre, who is logged in at it right now (the open session) and its ideal cycle time. This is the ordinary Manufacturing work centre with three fields added.



2 One work centre

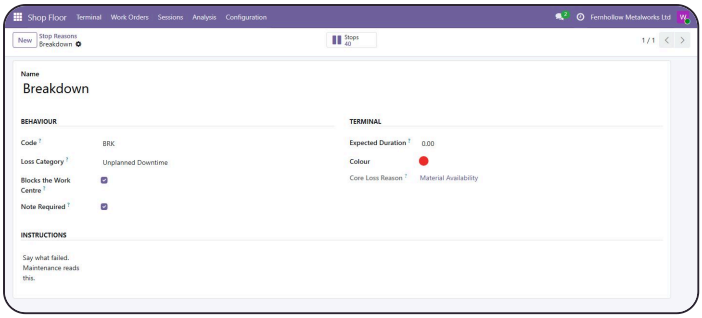
Ideal cycle time is the minutes the fastest possible unit takes on this machine; OEE performance is measured against it. **Stop reasons offered** limits the pause buttons on this terminal; left empty, every reason of the company is offered. Open Terminal opens the touch screen on this machine, Sessions lists its shifts.

The laser offers Nozzle and Lens Change, the weld bay Gas Cylinder Change. A press operator has no use for a button saying the oven is cold.



3 Stop reasons

Ordinary records: rename them, add your own, archive the ones you do not use. Eight ship with the module. The list shows the loss category and how many stops were logged.

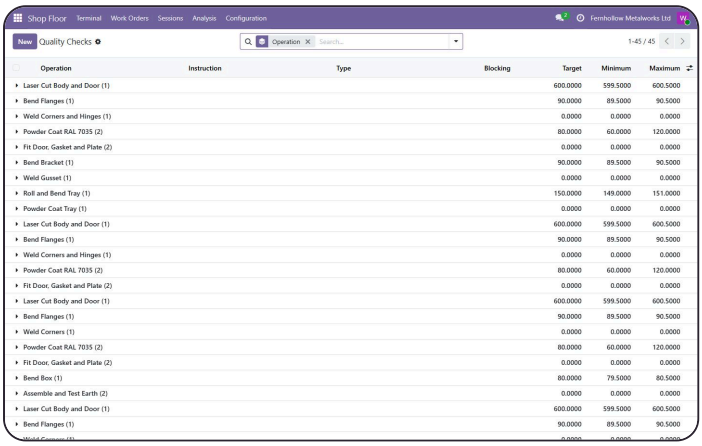


4

A stop reason

Loss category puts the minutes into unplanned downtime, reduced speed or quality loss. **Blocks the work centre** pauses the whole session, not just the job. **Note required** refuses the pause until the operator types what happened. **Expected duration** records the minutes a stop normally lasts, and **Instructions** holds a note for the operator. The core loss reason is filled in on first use, and the Stops button lists every pause on this reason.

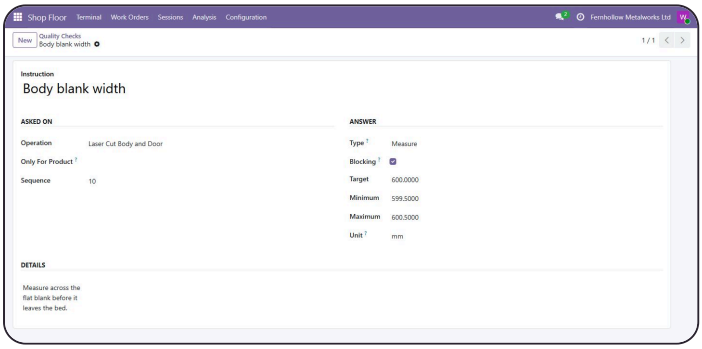
In this version nothing reads the expected duration, and the terminal does not show the instructions: both are reference fields for now.



5

Quality checks

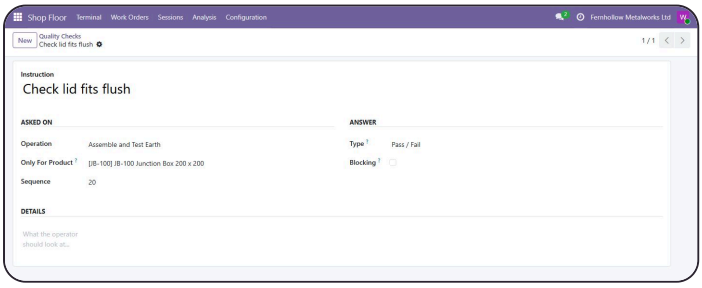
Each check is defined once, on an operation. Filters for blocking, measures and archived checks; group by operation or type.



6

A measure

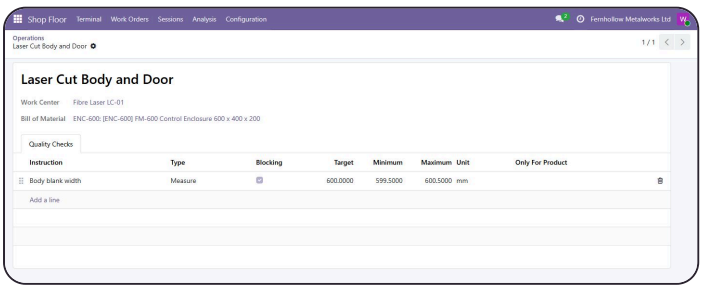
Target, minimum, maximum and a unit. The target has to sit inside its own range and the minimum below the maximum, or the record is refused. **Blocking** means the operation cannot be finished until this is answered without a failure.



7

Only for one product

Set **Only For Product** and the check is copied only onto work orders for that product. Here a lid check on a shared assembly operation, and it is not blocking: it is recorded but never stops the line.



8

Operations

The Operations screen lists every routing operation with its check count. Its Quality Checks tab is the same list, seen from the operation. Every work order created for this operation gets its own copy of these checks to answer, so a later change to the definition never rewrites an answer already given.

SHIPPED STOP REASON	LOSS CATEGORY	BLOCKING	NOTE	EXPECTED
Breakdown	Unplanned downtime	Yes	Required	
Setup and Changeover	Unplanned downtime			30 min
Waiting for Material	Unplanned downtime			
Break	Unplanned downtime			20 min
Minor Stop	Reduced speed			5 min
Running Slow	Reduced speed			
Rework	Quality loss		Required	
Quality Hold	Quality loss	Yes	Required	

PART 4 The terminal, step by step SHOP FLOOR

One screen per machine: the work orders waiting on it on the left, the job on the right, buttons big enough to press with a glove on. The screen only draws buttons. Every rule runs on the server.

IN THIS PART YOU WILL LEARN

- Opening the terminal on a machine and logging in
- Start, pause with a reason, resume, record, finish
- Answering the three kinds of check
- What happens when the network drops

TRY IT YOURSELF ON THE DEMO

1. Configuration > Terminals > MIG Weld Bay W-01 > Open Terminal
2. Pick the first work order: the weld check has failed and the job is held
3. Press Pause to see the reasons this machine offers (then leave it: nothing is sent until you pick one)

Open the terminal from the work centre. In this build, choosing a machine in the drop-down at the top of Shop Floor > Terminal raises a client error, because the template calls a function it cannot see. Opening it with **Open Terminal** on the work centre (or on a running session) presets the machine and works.



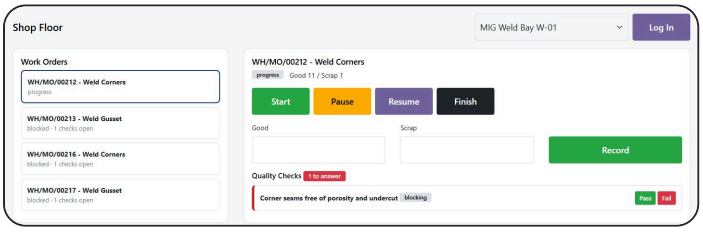
1 The terminal

Shop Floor > Terminal opens on a machine picker and a Log In button.



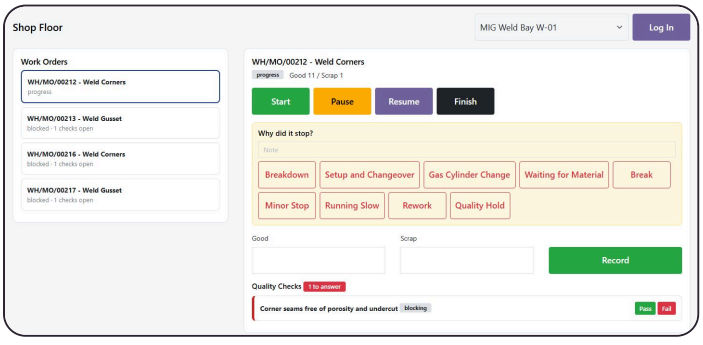
2 A machine

Opened from the work centre, the left column lists every work order on this machine that is not finished or cancelled, with its state and how many checks are still open.



3 A job

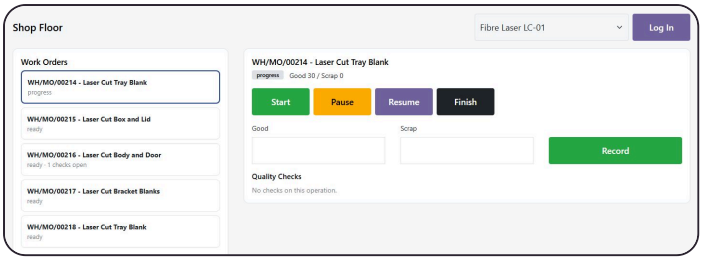
Pick a work order. The panel shows its state, the good and scrap recorded so far, the four big buttons, the quantity boxes and the quality checks, with a red badge counting the blocking ones still to answer.



4 Pause

Pause asks why it stopped. The buttons are the stop reasons this work centre offers. A reason that needs a note is refused until something is typed. A blocking reason also pauses the whole session.

The minutes from now until Resume are logged against that reason's loss, so they land in availability, performance or quality.



5

A running job

On the laser the job is running: the clock is on productive time. Type what came off the machine in Good and Scrap and press Record, as often as the shift needs.

What each button does on the server

BUTTON	WHAT IS WRITTEN	WHAT IS REFUSED
Log In	A session for this user on this machine, opened now	A machine that already has an open session
Start	Closes anything still running on the job, opens a productive time log, state In Progress	A finished or cancelled job
Pause	Closes the productive log, opens one against the reason's loss	No reason; no note where one is required; a job already paused
Resume	Closes the stop log, opens a productive one	A job that is not paused
Record	A production event with good and scrap	Negative or empty quantities, text that is not a number, a finished job
Pass / Fail / reading / text	The answer on the check, decided on the server	A fail with no note; a measure with no number; empty text
Finish	Closes the log, puts the good count on the work order (never more than is outstanding), finishes it	Any blocking check unanswered or failed. The message names them.
Log Out	Closes the session and anything left running on it	

When the network drops

Before it sends anything, the terminal makes a random request token and keeps sending the same one until the server answers. Every token is stored with a unique database index. A token seen before is a replay: the stored answer is handed back and not one quantity, minute or state is touched. The event records how many retries it absorbed, so a shift with a bad network shows up in the analysis instead of silently doubling its output.

The terminal sends the time the button was pressed, so an action queued while offline lands at the right minute. A time more than five minutes ahead of the server clock is not trusted and the server time is used instead.

> The terminal logs in the Odoo user who is signed in on the browser. There is no operator picker or badge login: each operator signs in to Odoo, or a shared terminal user is used per machine.

PART 5 Work orders, sessions and the event trail SHOP FLOOR

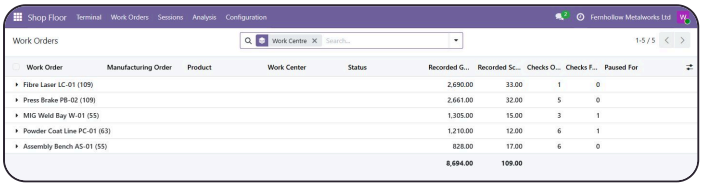
Everything the terminal does can be read back in the office: per job, per shift and per press of a button.

IN THIS PART YOU WILL LEARN

- The shop floor work order list and form
- Sessions: draft, running, paused and closed
- Running and stopped minutes on a shift
- Terminal events and retries absorbed
- The job that closes forgotten sessions

TRY IT YOURSELF ON THE DEMO

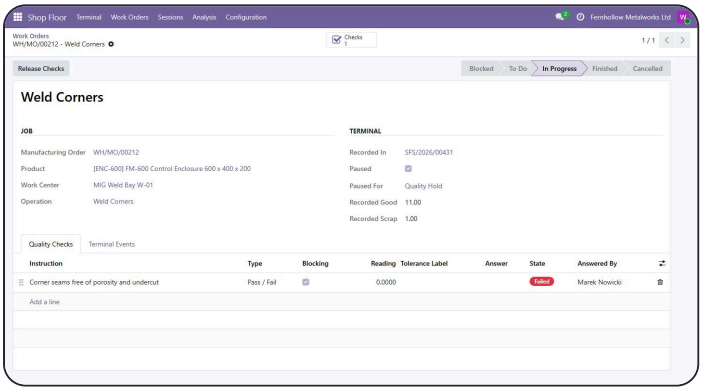
1. Work Orders: the yellow row is paused, the red one has a failed check
2. Sessions, switch to kanban: what is running right now
3. Analysis > Terminal Events, filter Retries Absorbed



1

Work orders

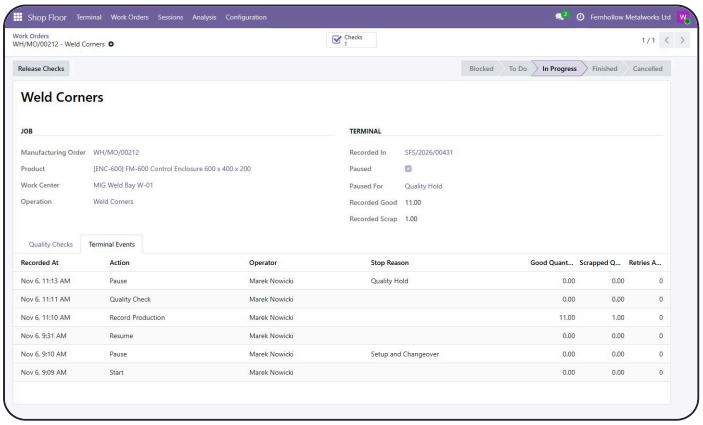
The shop floor's own list of work orders: manufacturing order, product, work centre, state, recorded good and scrap, checks open and failed, and the stop reason. Paused jobs are yellow, jobs with a failed check red. Filters for Paused, Checks Outstanding and Checks Failed.



2

A held job

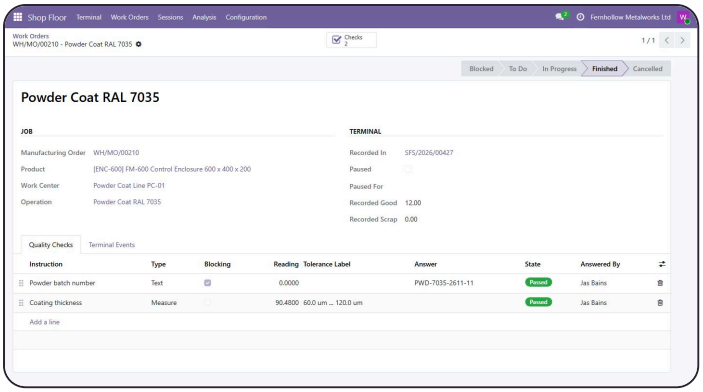
The Terminal block shows which session recorded it, the stop reason and the counts. The Quality Checks tab holds the answers, here a failed weld inspection. A supervisor sees **Release Checks** (see Part 6).



3

Its events

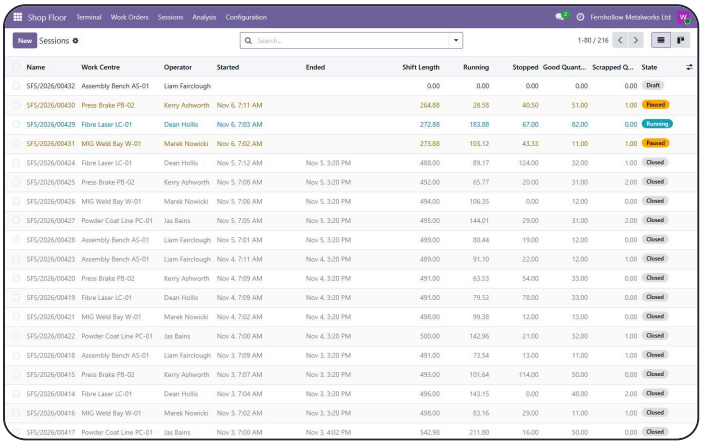
The Terminal Events tab lists every press on this job in order, with the operator, reason, quantities and retries absorbed.



4

A finished job

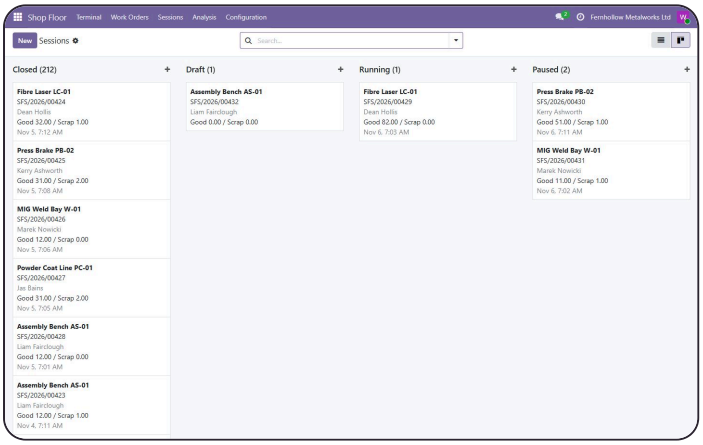
A finished powder coat operation: the batch number answered as text and the coating thickness recorded as a non-blocking measure.



5

Sessions

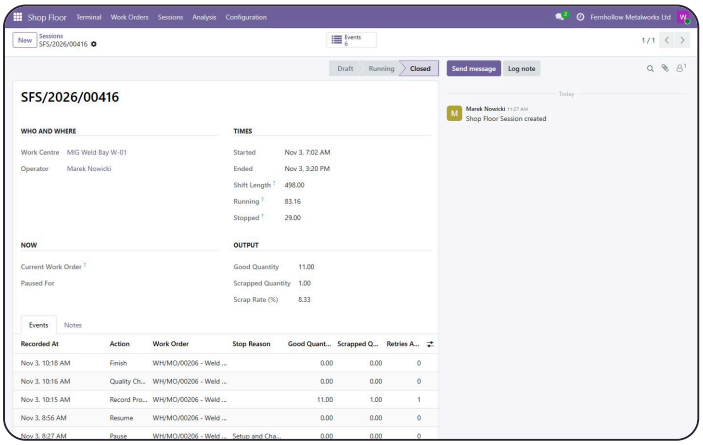
One row per operator per machine per shift: started, ended, state, good, scrap and scrap rate. Filters for Running, Paused, Closed, My Sessions and Started Today.



6

The floor right now

The kanban groups sessions by state. Running and Paused are the machines with someone on them; Draft is a session set up for a later shift.

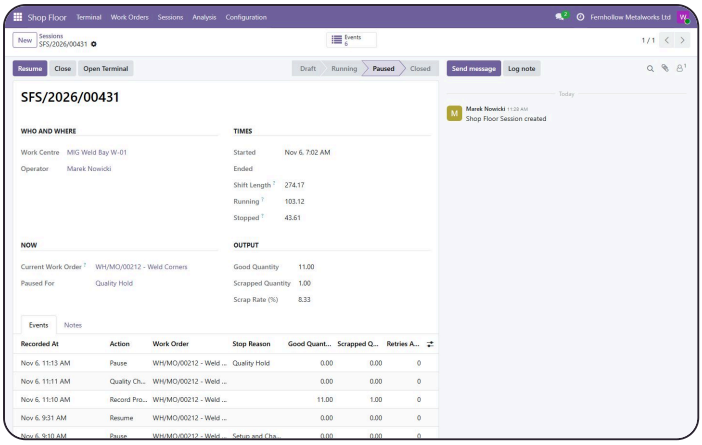


7

A closed shift

Shift length, running and stopped minutes, good, scrap and scrap rate. Running and stopped count every machine time log on this work centre inside the session window, clipped to it, so a break and a log that was already running when the shift began are both counted correctly.

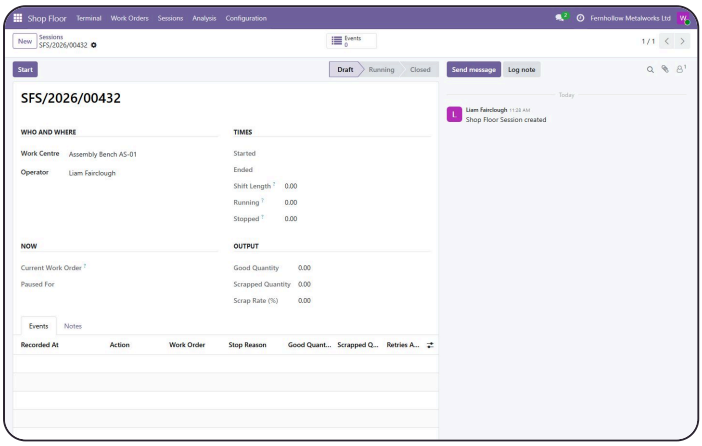
Only one session can be open on a machine at a time, so no two sessions can claim the same minute.



8

A paused shift

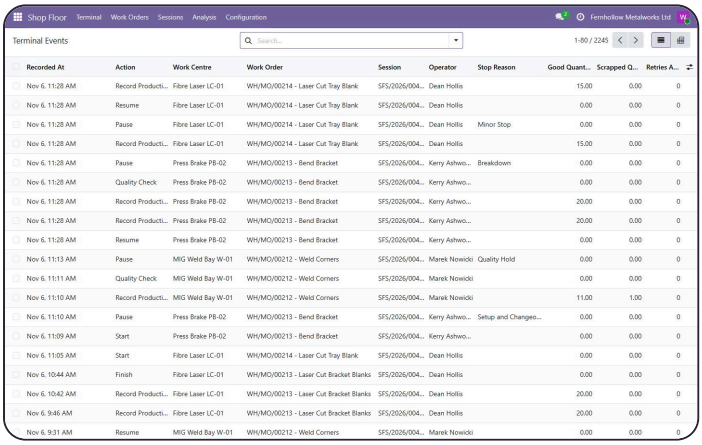
A blocking stop (here the quality hold on the weld bay) pauses the session as well as the job. The session form has Resume, Close and Open Terminal.



9

A draft session

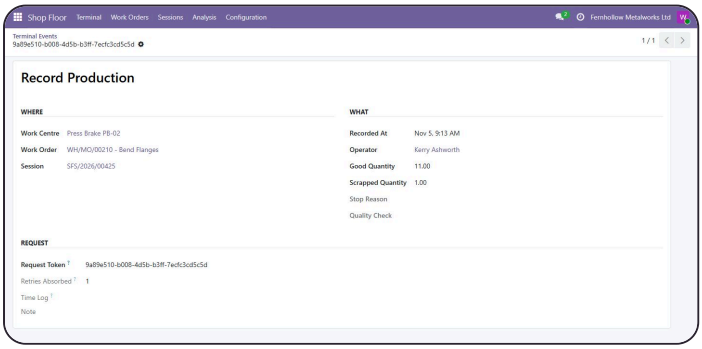
Created ahead of time for the afternoon cover. Start opens it.



10

Terminal events

Every press from every terminal. Rows with a retry absorbed are highlighted. Filters for Production, Stops, Retries Absorbed and Today; group by work centre, action, stop reason, operator or day. There is a pivot view as well.



11

One event

Where, what and the request token. Retries absorbed is 1: the terminal sent this production twice and the second one changed nothing.

Forgotten sessions

The scheduled action **Shop Floor: close forgotten sessions** runs every hour and closes any session still open or paused sixteen hours after it started, with a note in its chatter. A terminal left on over a weekend would otherwise keep the machine marked busy and stop the next shift logging in.

The session model has its own Pause for an operator break (logged against the work centre with no job on it, so it counts against availability without inflating one job). No button in the terminal or the session form calls it in this version: a break is paused on the job instead.

PART 6 Quality checks that hold the line SHOP FLOOR

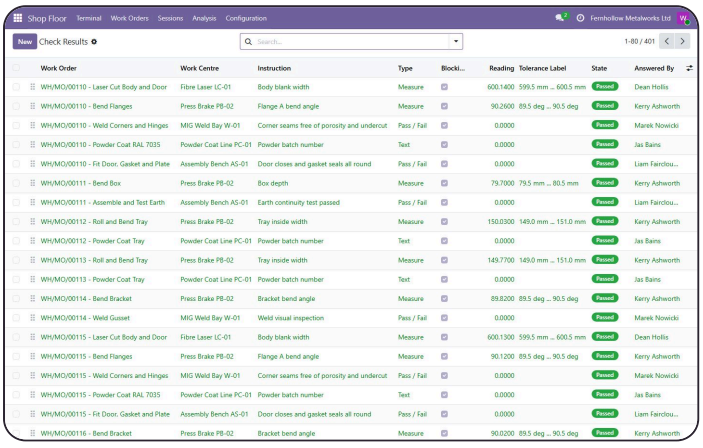
A check is asked on the terminal, decided on the server and kept on the work order. A blocking check stops the operation finishing until it is answered without a failure.

IN THIS PART YOU WILL LEARN

- Pass/fail, measure and text, and how each is decided
- What a failure blocks and who can release it
- Passing, failing and resetting from the office

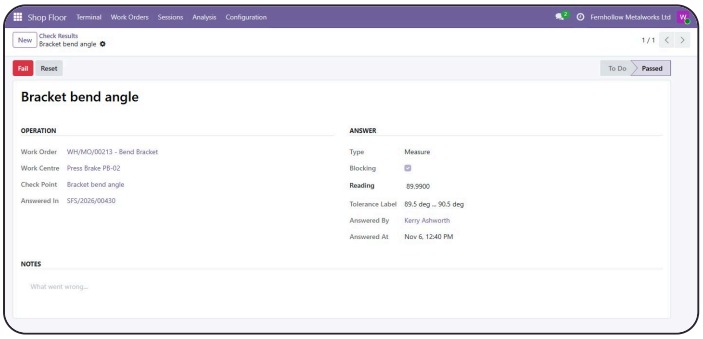
TRY IT YOURSELF ON THE DEMO

1. Analysis > Check Results, filter Failed
2. Open a passed measure and read its reading against the range



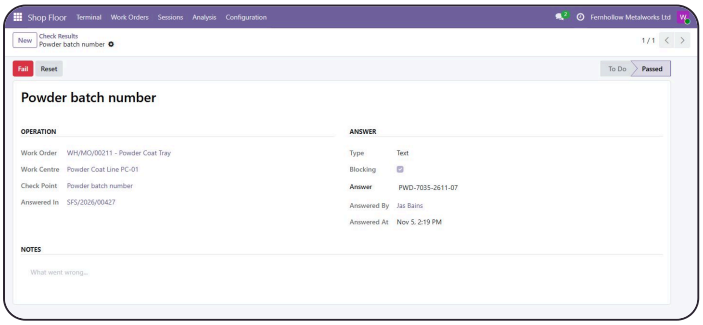
1 Check results

Every answer from every work order: instruction, type, blocking, reading or answer, result, who answered and when. Filters Outstanding, Failed, Blocking and Answered Today; group by work centre, result or work order.



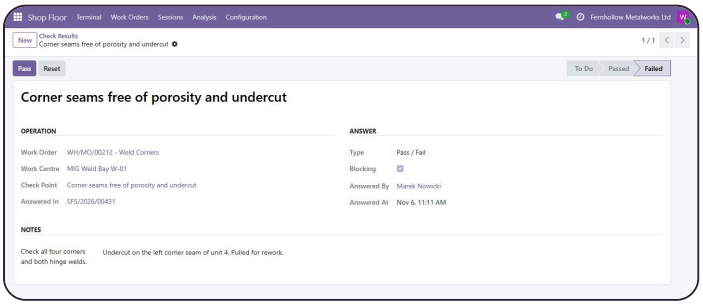
2 A measure

The reading is compared with the minimum and maximum on the server, with a tiny tolerance for gauge rounding. In range is Passed, out of range is Failed with the reading written into What Went Wrong. A measure cannot be set to Passed by hand, by import or through the API with a reading outside its range: the database constraint refuses it.



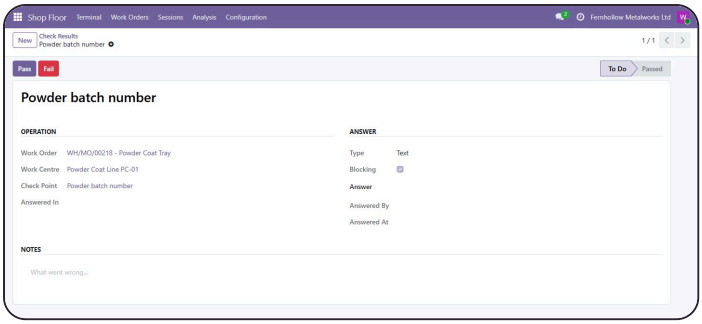
3 A text answer

A batch number, a serial label. Any non-empty answer passes.



4 A failure

A pass/fail check answered Fail needs a note saying what went wrong. On a blocking check the operation cannot be finished.



5

Not yet answered

Checks still To Do on jobs that have not reached the machine.

Getting past a failed check

WHO	HOW	WHAT IS KEPT
The operator	Re-measure or re-inspect and answer the check again on the terminal	The new answer on the check; the earlier attempt stays in the event trail
A shop floor manager	**Release Checks** on the work order (with a confirmation)	The operation finishes with checks open; the release is written to the server log
A shop floor manager	**Reset** on a failed check	The answer is cleared so it has to be taken again. Only managers can reset a failure; nobody can once the operation is finished

From the office a check result form also has **Pass** (not for measures, and a text check needs its answer first) and **Fail**.

PART 7 Measuring OEE SHOP FLOOR

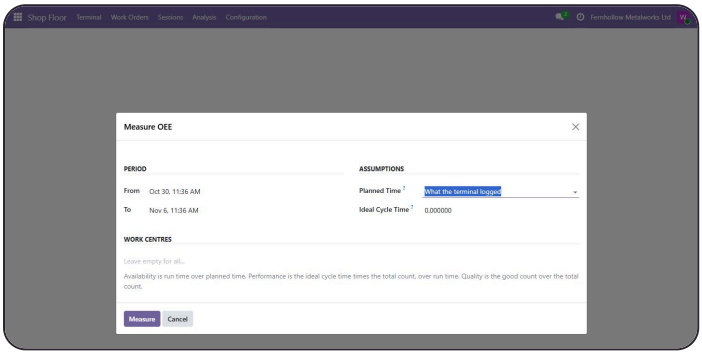
OEE is worked out on demand for a period and a set of machines, and every figure that went into it is kept on the record, so a number anybody disagrees with can be taken apart.

IN THIS PART YOU WILL LEARN

- The Measure OEE dialog and its three ways to set planned time
- Reading an OEE record
- Why performance is capped

TRY IT YOURSELF ON THE DEMO

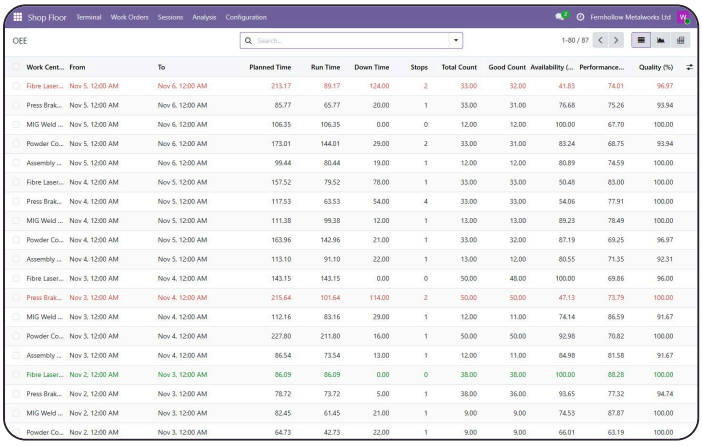
1. Analysis > OEE, open the weld bay for yesterday
2. Open the two records with a note: whole period and typed planned time



1

Measure OEE

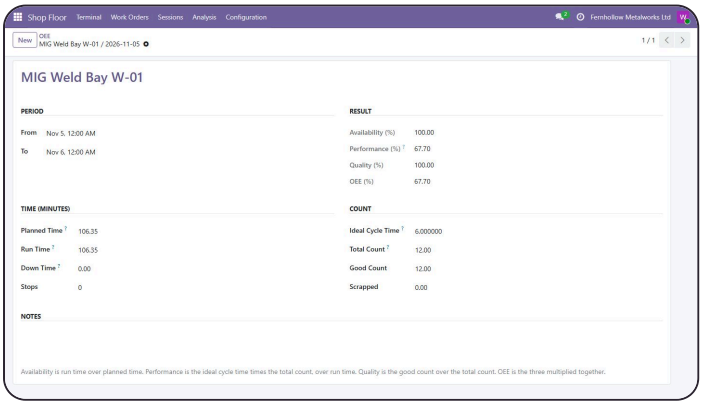
Analysis > Measure OEE (managers). Pick the period and the work centres (empty means all), how planned time is set and, optionally, an ideal cycle time to use instead of each machine's own. Measure creates one OEE record per work centre and opens them.



2

OEE

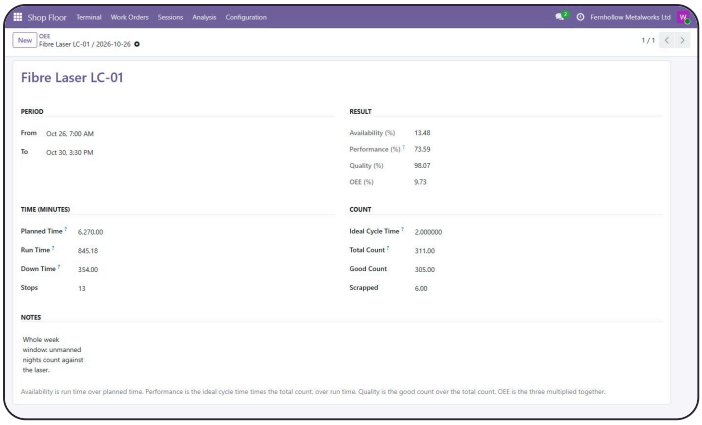
Every measurement with its period and the four percentages. Filters Below 40% and 85% and Above; group by work centre or period. There are graph and pivot views too.



3

One measurement

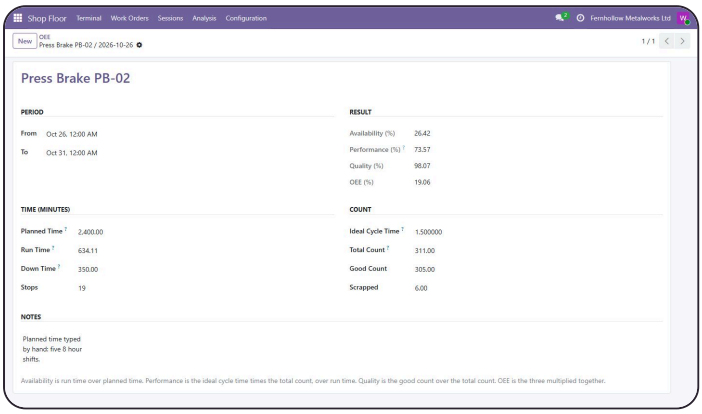
Planned, run and down time in minutes, the number of stops, the ideal cycle time, total, good and scrapped counts, then availability, performance, quality and OEE. Time logs are clipped to the period, so a night shift across midnight is not counted in two days.



4

The whole period

Planned time = the whole window. Every hour nobody was logged in counts against the machine, so availability drops sharply. Useful for a capacity view, not a shift one.



5

A typed planned time

Planned time typed by hand, here five eight hour shifts.

PLANNED TIME OPTION	AVAILABILITY IS MEASURED AGAINST
What the terminal logged (default)	Running plus stopped minutes the terminals recorded. Makes no claim about hours nobody logged.
The whole period	Every minute of the window
A figure I will type	The minutes typed; must be above zero

Performance is capped at 100% and the uncapped figure is kept beside it. A machine that beats its ideal cycle time means the ideal cycle time is wrong. The graph and pivot views add the percentages together when several records fall in one cell, so read them one period at a time.

PART 8 Change types, stages and the board PLM

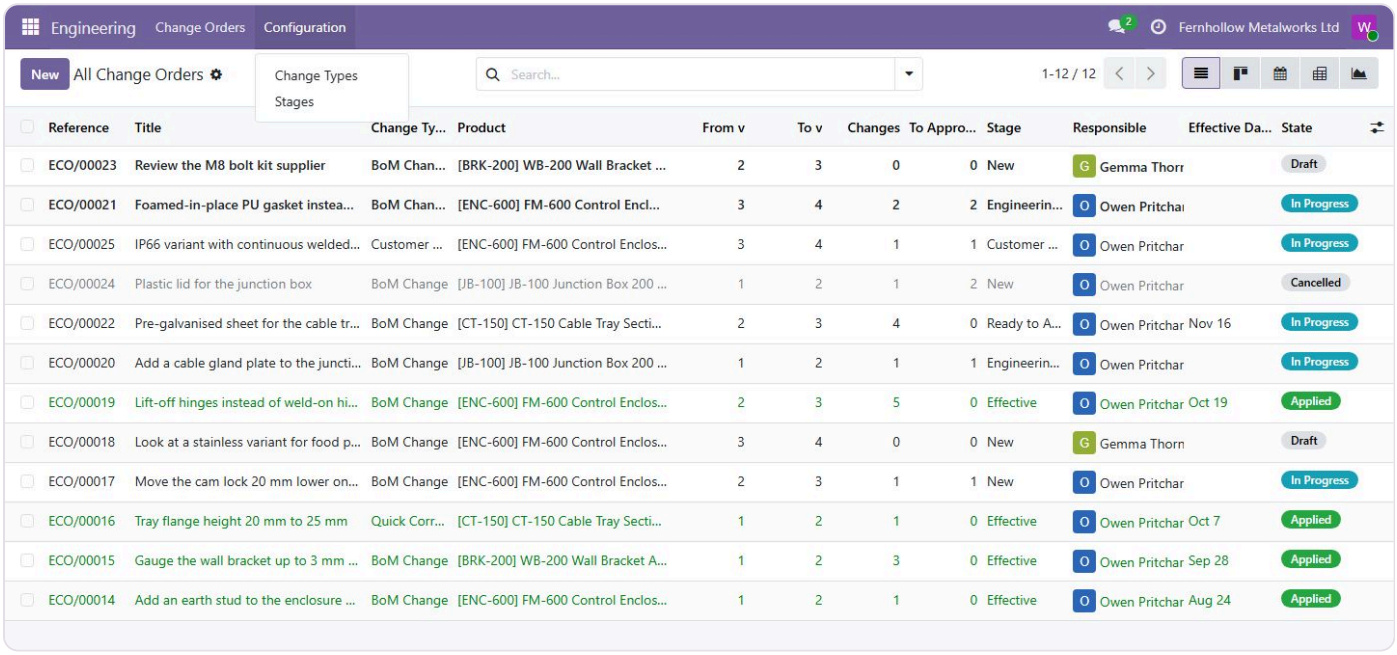
A change type is a board. Each board owns its columns, and the approval rule sits on the column, not on each change order, so it is written once.

IN THIS PART YOU WILL LEARN

- Change types and their stages
- Approval required, approvers and the effective stage
- Reading the change board

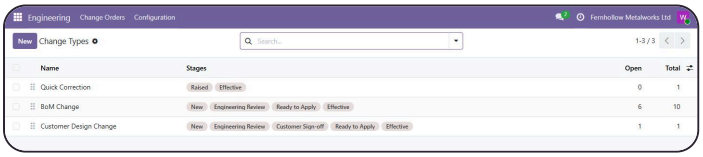
TRY IT YOURSELF ON THE DEMO

1. Engineering > Configuration > Change Types > Customer Design Change
2. Engineering > Change Orders > Change Board



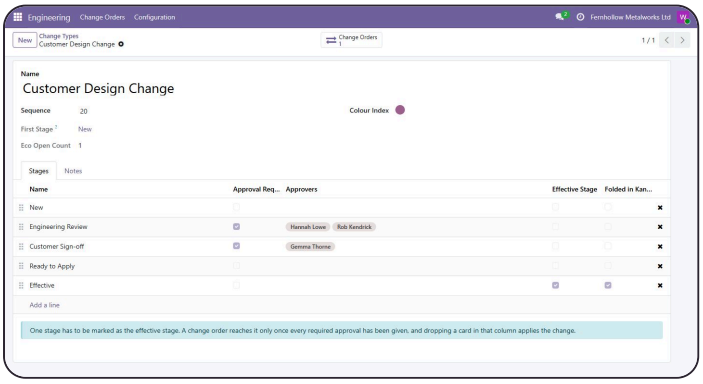
Reference	Title	Change Ty...	Product	From v	To v	Changes	To Appro...	Stage	Responsible	Effective Da...	State
ECO/00023	Review the M8 bolt kit supplier	BoM Chan...	[BRK-200] WB-200 Wall Bracket ...	2	3	0	0	New	Gemma Thorr		Draft
ECO/00021	Foamed-in-place PU gasket instea...	BoM Chan...	[ENC-600] FM-600 Control Encl...	3	4	2	2	Engineerin...	Owen Pritchai		In Progress
ECO/00025	IP66 variant with continuous welded...	Customer ...	[ENC-600] FM-600 Control Enclos...	3	4	1	1	Customer ...	Owen Pritchar		In Progress
ECO/00024	Plastic lid for the junction box	BoM Change	[JB-100] JB-100 Junction Box 200 ...	1	2	1	2	New	Owen Pritchar		Cancelled
ECO/00022	Pre-galvanised sheet for the cable tr...	BoM Change	[CT-150] CT-150 Cable Tray Secti...	2	3	4	0	Ready to A...	Owen Pritchar	Nov 16	In Progress
ECO/00020	Add a cable gland plate to the juncti...	BoM Change	[JB-100] JB-100 Junction Box 200 ...	1	2	1	1	Engineerin...	Owen Pritchar		In Progress
ECO/00019	Lift-off hinges instead of weld-on hi...	BoM Change	[ENC-600] FM-600 Control Enclos...	2	3	5	0	Effective	Owen Pritchar	Oct 19	Applied
ECO/00018	Look at a stainless variant for food p...	BoM Change	[ENC-600] FM-600 Control Enclos...	3	4	0	0	New	Gemma Thorn		Draft
ECO/00017	Move the cam lock 20 mm lower on...	BoM Change	[ENC-600] FM-600 Control Enclos...	2	3	1	1	New	Owen Pritchar		In Progress
ECO/00016	Tray flange height 20 mm to 25 mm	Quick Corr...	[CT-150] CT-150 Cable Tray Secti...	1	2	1	0	Effective	Owen Pritchar	Oct 7	Applied
ECO/00015	Gauge the wall bracket up to 3 mm ...	BoM Change	[BRK-200] WB-200 Wall Bracket A...	1	2	3	0	Effective	Owen Pritchar	Sep 28	Applied
ECO/00014	Add an earth stud to the enclosure ...	BoM Change	[ENC-600] FM-600 Control Enclos...	1	2	1	0	Effective	Owen Pritchar	Aug 24	Applied

Engineering > Configuration: Change Types and Stages.



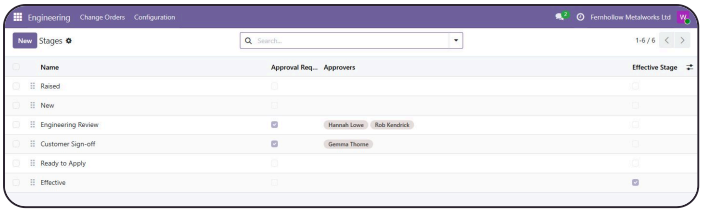
1 Change types

The module ships one, BoM Change. The demo adds Quick Correction (two columns, no approval) and Customer Design Change (five columns with a customer sign-off).



2 One type

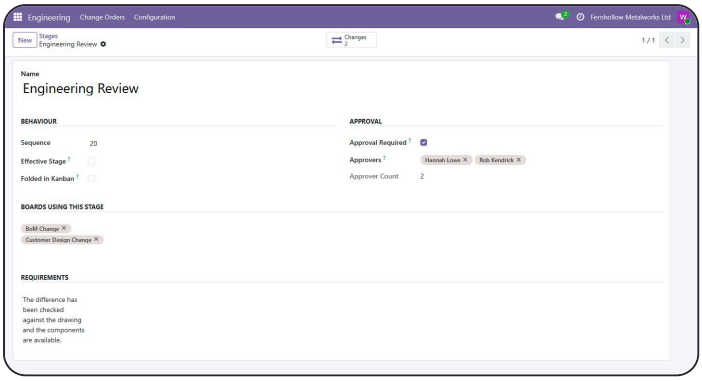
The stages in order. The first non-effective stage is where a new change order starts. A type with stages but no effective stage is refused, because nothing could ever be applied. The button lists the change orders of this type, and the counts show how many are open.



3

Stages

Filters for Needs Approval and Effective Stage. A stage with no company is shared by every company.

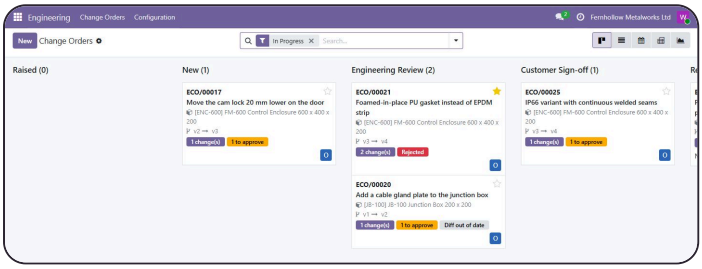


4

A stage with approvers

Approval Required with the approvers listed. Each approver gets their own line on every change order of a type that uses the stage. **Requirements** is shown on the board column. **Folded** collapses the column. **Effective Stage** marks the column a change order lands in once applied; it cannot ask for its own approval.

A stage that asks for approval with nobody listed is refused.



5

The change board

Kanban by stage, filtered to work in progress. A card shows the reference, title, product, the version change (v3 to v4), how many lines changed, and badges for a rejection or approvals still to give. Urgent ones carry a star.

SHIPPED STAGE	ORDER	WHAT IT MEANS
New	10	Raised. Open the revision and edit it.
Engineering Review	20	Approval required. Checked against the drawing, components available.
Ready to Apply	30	Every approval is in. Press Apply.
Effective	40	Effective stage, folded. The revision is live and the old version archived.

PART 9 Raising a change and editing the revision PLM

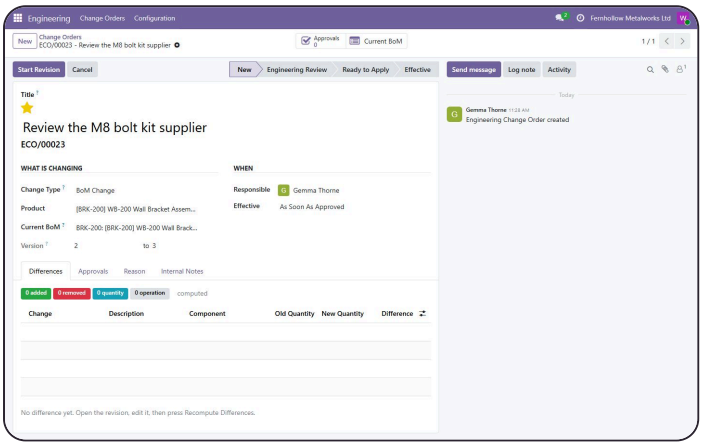
The change order takes an archived copy of the live bill of materials. The engineer edits the copy. The difference between the two is what gets reviewed.

IN THIS PART YOU WILL LEARN

- Raising a change order against the live BoM
- Start Revision and the archived copy
- The difference: components, quantities, operations, the BoM itself
- The out of date warning

TRY IT YOURSELF ON THE DEMO

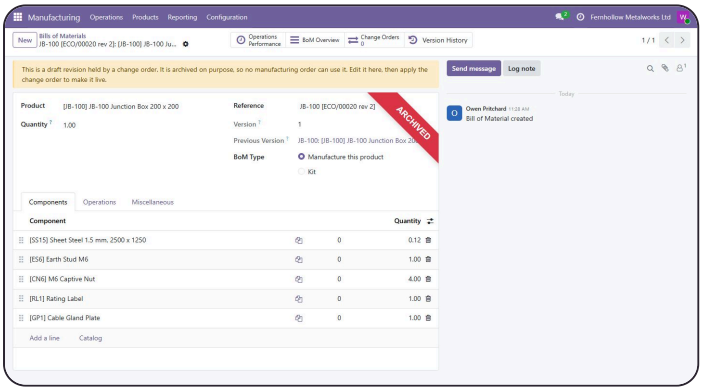
1. Open ECO/00023 (draft): no revision yet
2. Open ECO/00020 and read the Differences tab and the warning above it



1

Raise it

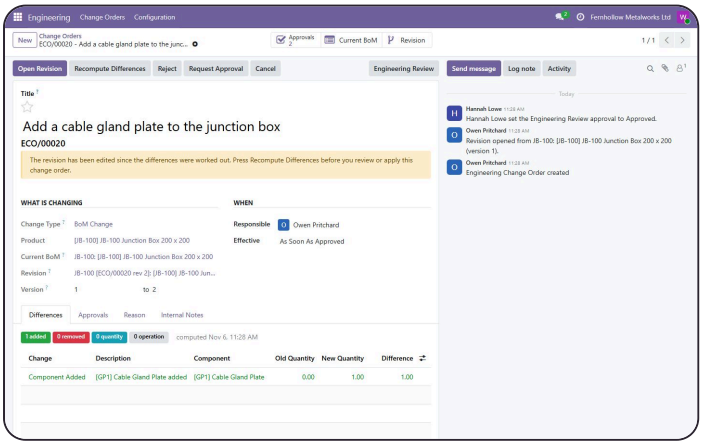
Title, change type, product and the current bill of materials (only live ones of that product, never a revision held by another change), the responsible engineer, the reason and internal notes, priority, and when it should take effect: as soon as approved, or at a date.



2

Start Revision

Copies the current BoM with its components and operations. The copy is **archived**, so no manufacturing order can pick up a half finished revision, and the form says so in a yellow banner. Edit it here like any bill of materials. The change order moves to In Progress and its approval lines are created from the stages.

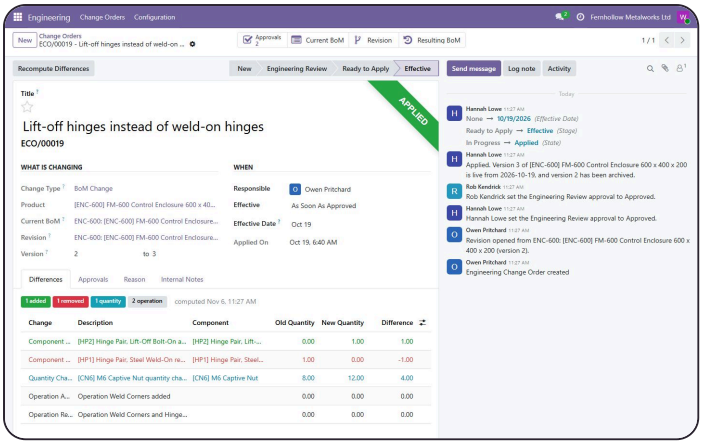


3

Differences

Worked out line by line and stored: components added and removed, quantities changed, operations added, removed or with a new duration, and changes to the BoM itself (quantity, unit, type). Counters at the top. **Recompute Differences** works them out again.

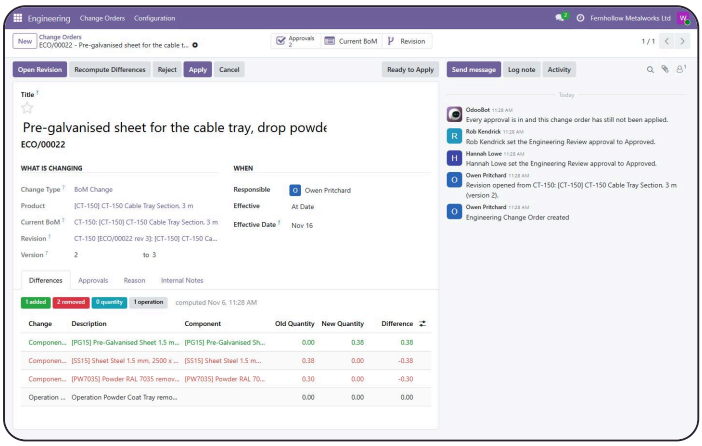
The yellow warning appears when the revision has been edited since the difference was worked out, as here.



4

A bigger change

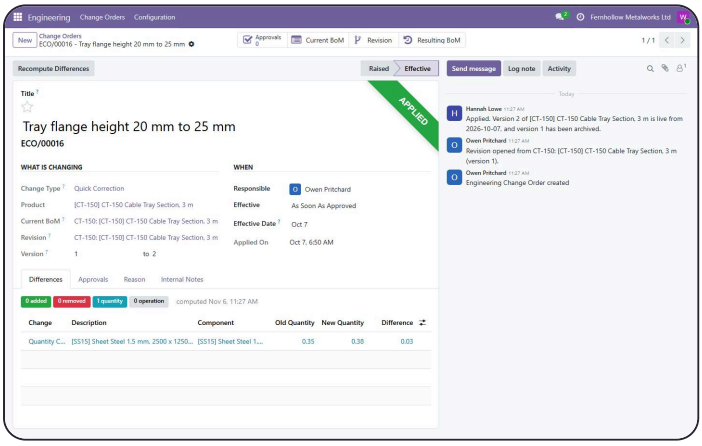
An applied change: a hinge swapped, a nut quantity raised and the weld operation retimed. The weld operation was also renamed, so it shows as one operation removed and one added: operations are matched by name and work centre.



5

Removing an operation

Pre-galvanised sheet instead of sheet plus powder: two components removed, one added and the powder coat operation taken out. This one is fully approved and set to take effect at a date.

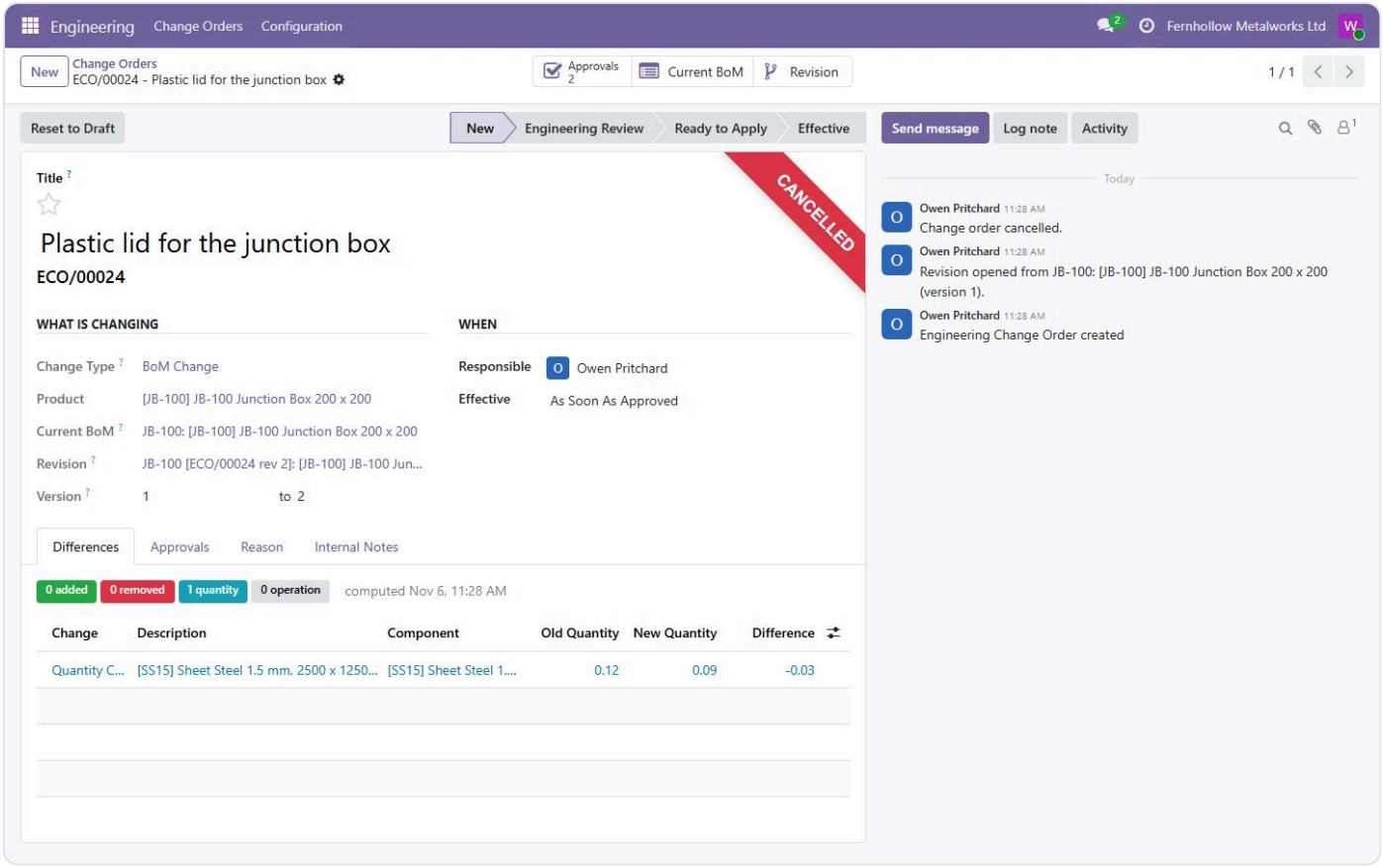


6

A quick correction

On the Quick Correction board there is no approval stage, so a change can be applied as soon as its revision is right.

Stat buttons on the change order open the **Current BoM**, the **Revision** and, once applied, the **Resulting BoM**. Cancel keeps the revision on the record, archived, so the work is not lost; Reset to Draft brings a cancelled change back (to In Progress if it already had a revision).



A cancelled change order, with its revision kept.

PART 10 Approvals PLM

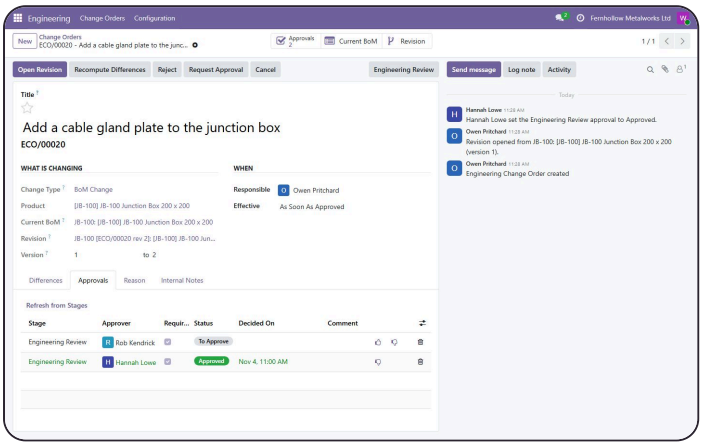
Approvals are rows, one per approver per stage, because the question asked most is who said yes and when.

IN THIS PART YOU WILL LEARN

- Approving, rejecting and changing your mind
- Request Approval and the email it sends
- My Approvals and All Approvals

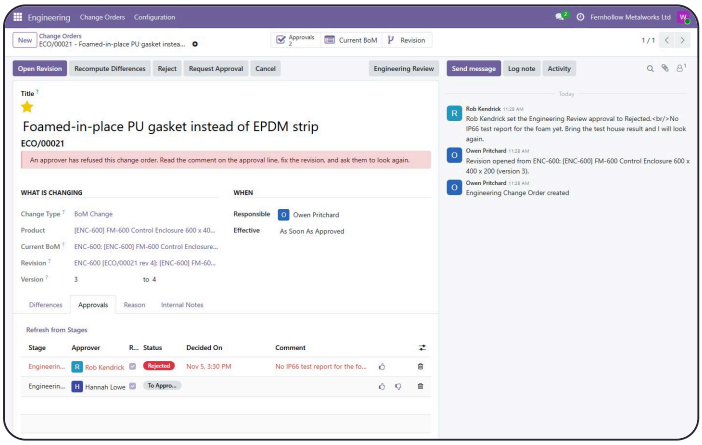
TRY IT YOURSELF ON THE DEMO

1. Open ECO/00021 and read the rejection
2. Change Orders > All Approvals, filter To Approve



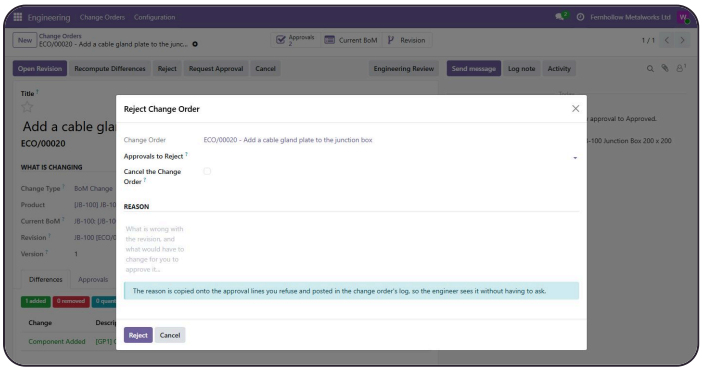
1 The approvals tab

One line per approver: stage, required, status, decided on and comment, with per-line approve and reject buttons. **Refresh from Stages** adds lines for approvers added to a stage and drops unanswered lines for people taken off; answers already given are kept.



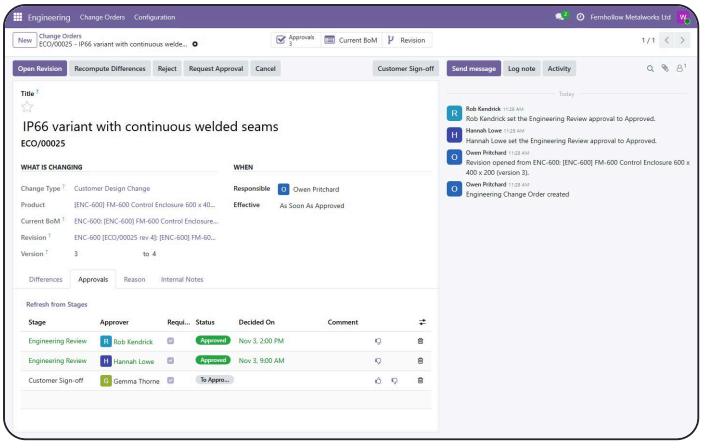
2 A rejection

Rejecting opens a dialog that asks for a reason. The reason goes on the approval line and into the chatter for the engineer, and a red banner on the change order says an approver has refused it. A rejected approval can later be approved by the same person.



3 The reject dialog

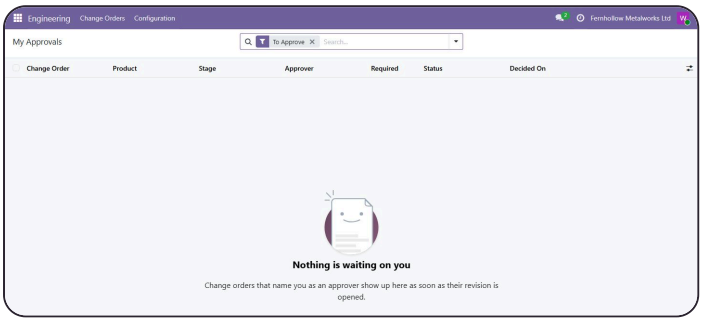
Pre-filled with your own approval lines. Tick **Cancel the change order** to stop it altogether instead of sending it back.



4

Customer sign-off

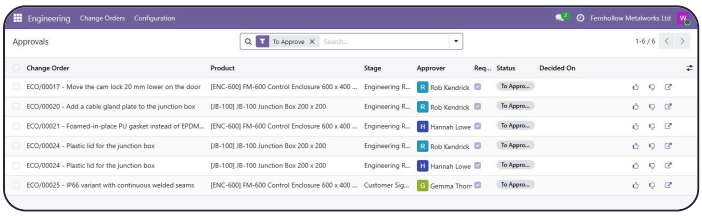
On the Customer Design Change board, engineering and quality have approved and the change waits for production to confirm the customer's signed drawing.



5

My Approvals

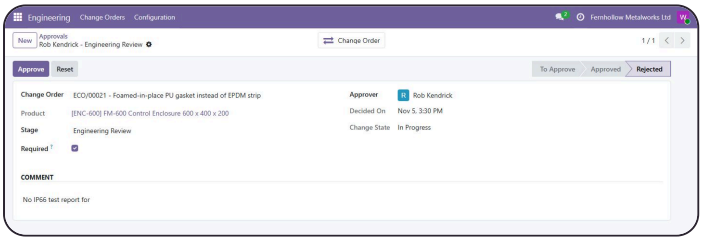
Every line waiting on the signed in user.



6

All Approvals

Configurators see every line. Filters Mine, To Approve, Approved, Rejected and Change Order Open; group by approver, stage, status or change order.



7

An approval

Approve, Reject, Reset (withdraws an answer) and a button to open the change order. Answers cannot change once the change order is applied or cancelled.

BUTTON ON THE CHANGE ORDER	WHO SEES IT	WHAT IT DOES
Approve	An approver with a line waiting	Approves every line on this change waiting on you
Reject	Anyone, while in progress	The reject dialog
Request Approval	While approvals are outstanding	One email to everyone whose answer is not yet an approval, rejected ones included, and a note in the chatter

Only the named approver can answer a line, unless the user has Configure Engineering Changes. The approval request uses the email template **ECO approval request**; outgoing mail has to be set up in Odoo for it to leave.

PART 11 Applying, versions and traceability PLM

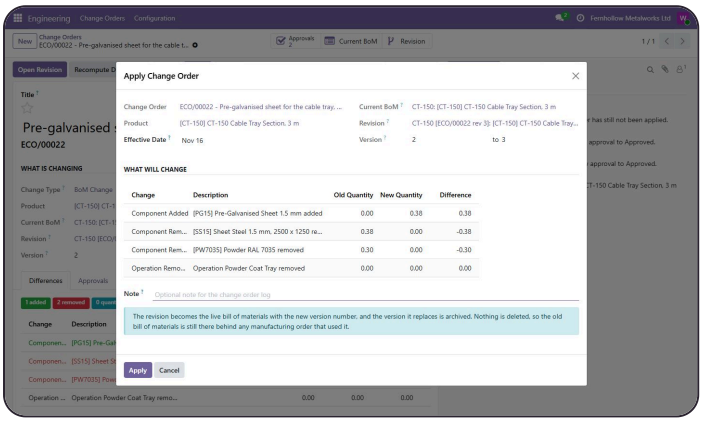
Applying swaps the revision in as the live bill of materials with the version number up by one. The version it replaces is archived, never deleted.

IN THIS PART YOU WILL LEARN

- The Apply dialog and the effective date
- Two changes on the same version
- Version history on the bill of materials
- The reminder job and the reporting views

TRY IT YOURSELF ON THE DEMO

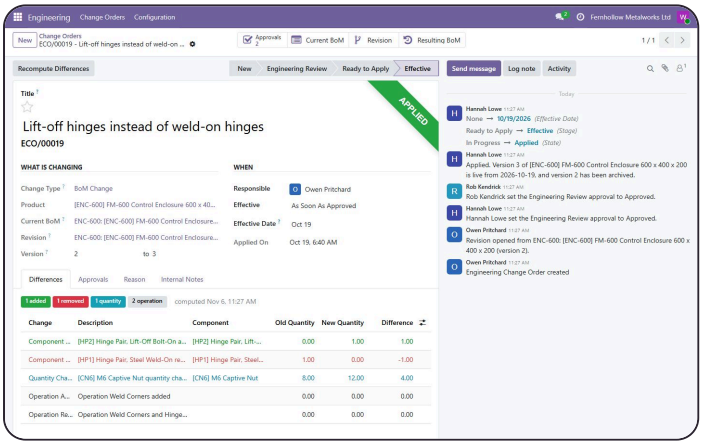
1. Open the live BoM of FM-600 and press Version History
2. Read the chatter of ECO/00017 and ECO/00018



1

Apply

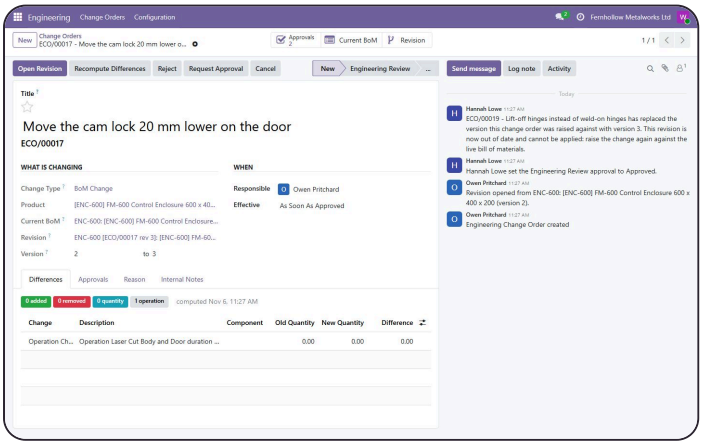
Shown once every required approval is in. The dialog lists the differences, the old and new version and the effective date to stamp (today by default), with a note for the chatter. It recomputes the difference first and refuses a revision identical to the live BoM.



2

Applied

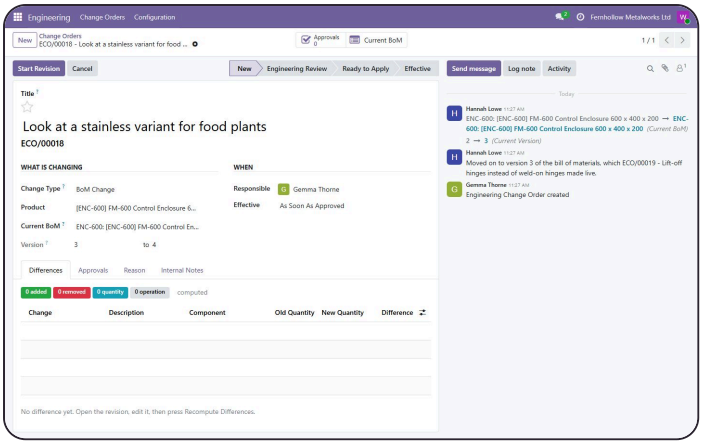
The revision becomes live as version 3, version 2 is archived, the effective date is stamped on the new BoM and the change lands in the Effective column. Dropping a card in the effective column does the same and is refused for the same reasons.



3

Overtaken

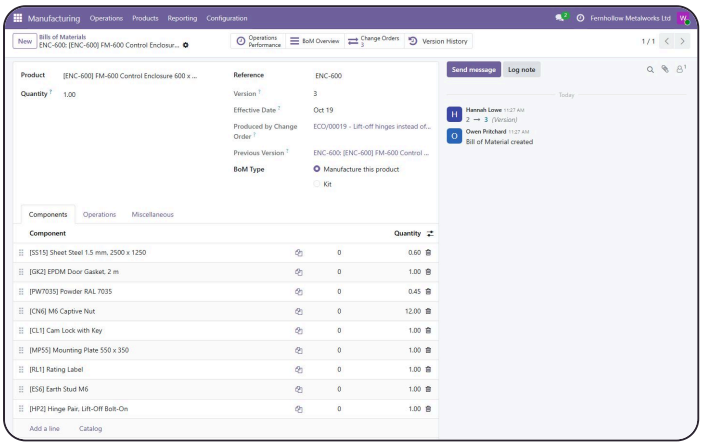
ECO/00017 was raised against version 2 and had opened its revision when ECO/00019 made version 3 live. It is told in its chatter that it is out of date, and Apply refuses it: applying it would quietly put version 2 back.



4

Moved on

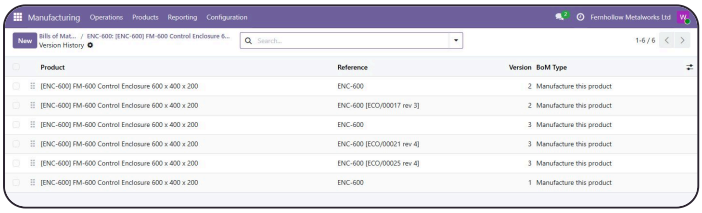
ECO/00018 was a draft with no revision yet, so it simply moved on to version 3, and its chatter says so.



5

The bill of materials

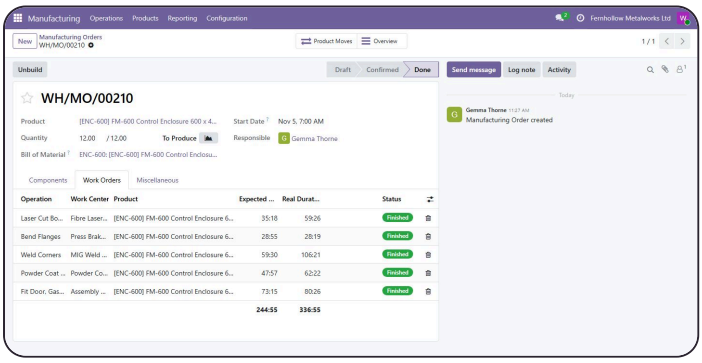
Version, effective date, the change order that produced it and the version it replaced. Stat buttons: **Change Orders** raised against it and **Version History**.



6

Version history

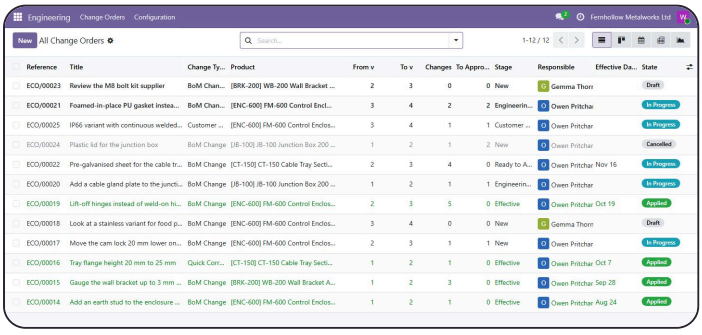
The whole chain, archived versions and open revisions included. The BoM list has a Version column and the filters Draft Revisions and From a Change Order.



7

What production used

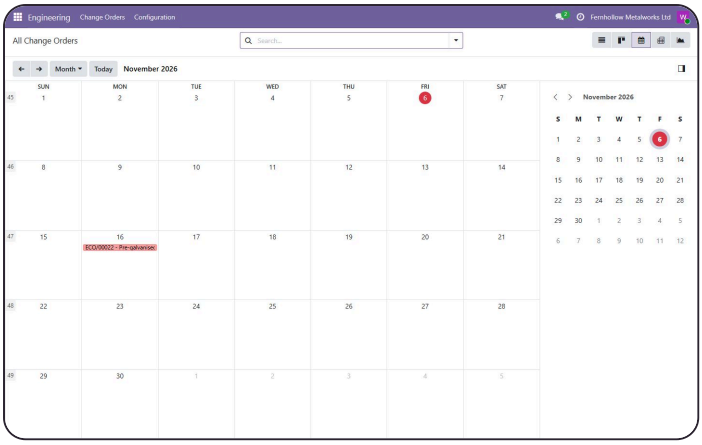
A finished manufacturing order keeps the bill of materials it was made from, so the version built on any day can be traced back.



8

All change orders

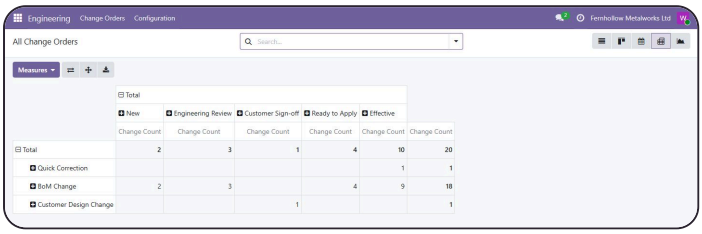
Every change order with list, kanban, calendar, pivot and graph views. Filters My Change Orders, Waiting on Me, Draft, In Progress, Fully Approved, Applied, Urgent and Past Effective Date.



9

Calendar

Change orders on their effective dates.



10

Pivot

The number of differences by change type and stage. Components added, removed, quantity and operation changes are stored too and can be picked as measures.

The reminder

The scheduled action **Engineering Changes: chase what is sitting still** (shipped switched off) posts a note on a change order that is fully approved but not applied, or whose effective date has passed while approvals are still outstanding. It writes at most once a day per change order.

> Applying replaces the bill of materials; it does not touch manufacturing orders already confirmed on the old version.

PART 12 Printed documents

Two printouts: one for the supervisor at the end of a shift, one for the change file.



FERNHOLLOW

Fernhollow Metalworks Ltd
Unit 4, Canal Side Works
Mill Lane
Rotherham
S60 0XX
United Kingdom

Shift Report
SFS/2026/00416

Work Centre	MIG Weld Bay W-01	Shift Minutes	498.0
Operator	Marek Nowicki	Running	83.155164
Started	11/03/2026 07:02:00 AM	Stopped	29.0
Ended	11/03/2026 03:20:00 PM	Good / Scrap	11.00 / 1.00

What Happened

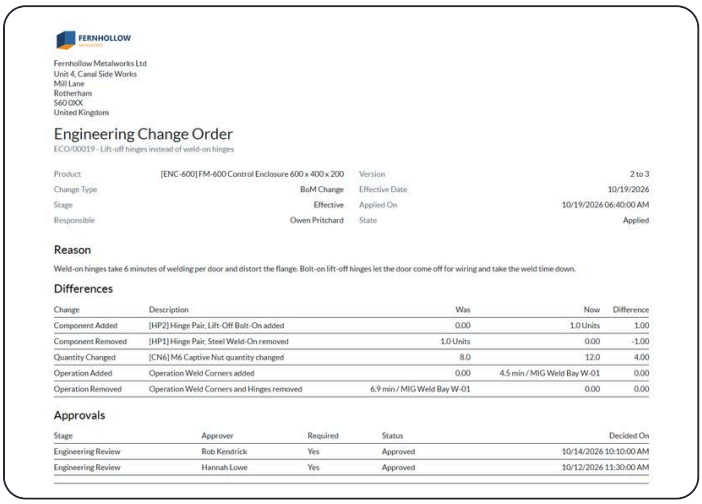
Time	Action	Work Order	Reason	Good	Scrap
11/03/2026 10:18:15 AM	Finish	WH/MO/00206 - Weld Corners		0.00	0.00
11/03/2026 10:16:15 AM	Quality Check	WH/MO/00206 - Weld Corners		0.00	0.00
11/03/2026 10:15:15 AM	Record Production	WH/MO/00206 - Weld Corners		11.00	1.00
11/03/2026 08:56:06 AM	Resume	WH/MO/00206 - Weld Corners		0.00	0.00
11/03/2026 08:27:06 AM	Pause	WH/MO/00206 - Weld Corners	Setup and Changeover	0.00	0.00
11/03/2026 08:26:06 AM	Start	WH/MO/00206 - Weld Corners		0.00	0.00

1

Shift Report SHOP FLOOR

Print it from a session. Work centre, operator, start and end, shift minutes, running and stopped minutes, good and scrap, then every terminal event of the session in order, with the reason and quantities, and the session notes.

The running and stopped minutes print with every decimal of the stored figure.



FERNHOLLOW

Fernhollow Metalworks Ltd
Unit 4, Canal Side Works
Mill Lane
Rotherham
S60 0XX
United Kingdom

Engineering Change Order
ECO/00019 - Lift-off hinges instead of weld-on hinges

Product	[ENC-600] FM-600 Control Enclosure 600 x 400 x 200	Version	2 to 3
Change Type	BoM Change	Effective Date	10/19/2026
Stage	Effective	Applied On	10/19/2026 06:40:00 AM
Responsible	Owen Pritchard	State	Applied

Reason

Weld-on hinges take 6 minutes of welding per door and distort the flange. Bolt-on lift-off hinges let the door come off for wiring and take the weld time down.

Differences

Change	Description	Was	Now	Difference
Component Added	[HP2] Hinge Pair, Lift-Off Bolt-On added	0.00	1.0 Units	1.00
Component Removed	[HP1] Hinge Pair, Steel Weld-On removed	1.0 Units	0.00	-1.00
Quantity Changed	[CN6] M6 Captive Nut quantity changed	8.0	12.0	4.00
Operation Added	Operation Weld Corners added	0.00	4.5 min / MIG Weld Bay W-01	0.00
Operation Removed	Operation Weld Corners and Hinges removed	6.9 min / MIG Weld Bay W-01	0.00	0.00

Approvals

Stage	Approver	Required	Status	Decided On
Engineering Review	Rob Kendrick	Yes	Approved	10/14/2026 10:10:00 AM
Engineering Review	Hannah Lowe	Yes	Approved	10/12/2026 11:30:00 AM

2

Engineering Change Order PLM

Print it from a change order. Product, type, stage, responsible, the version change, effective date and applied on, the state, the reason, every difference with was, now and the difference, and the approvals with stage, approver, required, status, date and comment.



FERNHOLLOW

Fernhollow Metalworks Ltd
Unit 4, Canal Side Works
Mill Lane
Rotherham
S60 0XX
United Kingdom

Engineering Change Order
ECO/00021 - Foamed-in-place PU gasket instead of EPDM strip

Product	[ENC-600] FM-600 Control Enclosure 600 x 400 x 200	Version	3 to 4
Change Type	BoM Change	Effective Date	as soon as approved
Stage	Engineering Review	Applied On	not applied
Responsible	Owen Pritchard	State	In Progress

Reason

Hand-fitted EPDM strip is 4 minutes a door and the corner joints are where the leaks come from.

Differences

Change	Description	Was	Now	Difference
Component Removed	[GK2] EPDM Door Gasket, 2 m removed	1.0 Units	0.00	-1.00
Operation Changed	Operation Fit Door, Gasket and Plate duration changed	5.75 min	4.0 min	0.00

Approvals

Stage	Approver	Required	Status	Decided On
Engineering Review	Rob Kendrick	Yes	Rejected	11/05/2026 03:30:00 PM
Engineering Review	Hannah Lowe	Yes	To Approve	

3

A rejected change on paper PLM

The same document for a change still in review: the rejection and its comment are on the sheet.

PART 13 A fifteen minute walk-through

If you have the demo open and fifteen minutes, this path shows both modules.

#	DO THIS	AND NOTICE
1	Shop Floor > Sessions, kanban view	*Who is on which machine right now: one running, two paused, one draft.*
2	Configuration > Terminals > MIG Weld Bay W-01 > Open Terminal	*The weld job held on a failed seam check.*
3	Press Pause, look, then leave it	*The reasons this machine offers, including Gas Cylinder Change.*
4	Work Orders, open the red row	*The failed check and, for a supervisor, Release Checks.*
5	Its Terminal Events tab	*Every press on the job, in order.*
6	Analysis > Terminal Events, filter Retries Absorbed	*Requests sent twice and counted once.*
7	Analysis > OEE, open the weld bay for a week	*Availability, performance, quality and every figure behind them.*
8	Sessions, open a closed shift, Print > Shift Report	*The shift on one page.*
9	Engineering > Change Board	*Work in progress by stage, with rejected and waiting badges.*
10	Open ECO/00020, Differences tab	*The difference worked out, and the out of date warning.*
11	Open ECO/00021, Approvals tab	*Quality rejected it, with the reason.*
12	Open ECO/00019	*Applied: version 3 live, version 2 archived, the hinge swap and the weld retime.*
13	Open ECO/00017, read the chatter	*Overtaken by ECO/00019 and refused if applied.*
14	Manufacturing > Bills of Materials > FM-600 > Version History	*Versions 1 to 3 and the open revisions.*
15	Print ECO/00019	*The change order as it goes in the file.*

PART 14 Questions people ask

Short answers, all from what the modules actually do.

QUESTION	ANSWER
Do they need Odoo Enterprise?	No. Both run on Odoo 19 Community with Manufacturing.
Do I have to buy both?	No. Each works on its own. Together, checks travel with operations into new BoM versions.
Does the terminal connect to the machine?	No. Operators type good and scrap. There is no PLC or counter link.
What does it run on?	Any browser on a tablet or a PC by the machine. Buttons are sized for touch.
Can two people log in on one machine?	No. One open session per work centre, so minutes are never counted twice.
What if the tablet loses its network?	It resends the same request token; the server ignores a token it has seen.
Can an operator pass an out of tolerance part?	No. The server decides, and a passed measure outside its range is refused even through the API.
Can a supervisor override a check?	Yes, Release Checks on the work order, for shop floor managers. It is logged.
Are the Odoo OEE figures still right?	Yes. Every start, pause and resume writes a normal work centre time log.
Is the live BoM edited during a change?	No. The revision is an archived copy until Apply.
What happens to the old BoM?	Archived with its version number, never deleted, and reachable from Version History.
Can two changes on one version both be applied?	No. The second is refused and told why.
Can approvals differ per kind of change?	Yes. Each change type has its own stages, and approvers sit on the stage.
Does applying update confirmed manufacturing orders?	No. They keep the version they were confirmed with.

PART 15

The menu map

Every screen in both modules, on one page.

Shop Floor

Terminal	The touch screen per machine
Work Orders	Jobs with recorded good, scrap, checks and stop reason
Sessions	One operator, one machine, one shift

Shop Floor › Analysis

Measure OEE	Managers: work out OEE for a period
OEE	Every measurement, list, graph, pivot
Check Results	Every quality answer
Terminal Events	Every press, with retries absorbed

Shop Floor › Configuration

Terminals	Work centres: ideal cycle, stop reasons offered
Stop Reasons	Loss category, blocking, note required
Quality Checks	Check points per operation
Operations	Routing operations and their checks

Engineering

Change Board	Kanban of change orders by stage
All Change Orders	List, calendar, pivot, graph
My Approvals	Lines waiting on you
All Approvals	Configurators: every approval line

Engineering › Configuration

Change Types	One board per kind of change
Stages	Columns, approvers, effective stage

Manufacturing (added)

Bills of Materials	Version, effective date, change orders, version history
Work Centers	Open Terminal and Sessions on the work centre form

Want this on your shop floor?

We set up your work centres, stop reasons and quality checks, put the terminal on the tablets by each machine, build your change types and approval stages, load your bills of materials and train the operators and engineers.

way4tech.com · info@way4tech.com